Reasoning agent

Representation and inference

The agent accepts logical formulas directly or translates English into logic, then computes the answer symbolically. Lark handles formulas and controlled English rules, with LemmInflect for verb normalization. A small parser handles causal templates. Other English sentences go to a local Qwen semantic parser with XGrammar-constrained JSON output. The model receives the premises and query without labels or reference formulas; its output contains formulas rather than a final answer.

First-order formulas use prefix trees for predicates, Boolean operators, quantifiers and equality. Free premise variables are universally quantified. Z3 checks KB AND NOT(query): unsatisfiability establishes entailment. A second check, KB AND query, distinguishes refutation from an unknown result. Checks have a two-second timeout. An inconsistent classical knowledge base entails every query and is flagged separately.

RuleTaker uses a separate closed-world mode: ground Horn rules are evaluated by alternating lower and upper fixed points, with negation as failure. Only positive rule heads add positive support. Causal reasoning uses Boolean equations in factual and counterfactual worlds, sharing exogenous variables. Interventions replace equations. Cycles are rejected; unspecified roots default to True. Nondegenerate Bernoulli roots allow both values. The query must hold in every world consistent with the evidence. Contradictory causal evidence returns False with a diagnostic.

Model comparison

Three open-weight models were compared on the same 92 training examples: 48 FOLIO (16 per label), 24 RuleTaker (three per configuration), and 20 CounterBench (four per type). All used the same prompt, symbolic backend, BF16 precision, non-thinking decoding and seed 42. The comparison bypassed the controlled parsers. Macro accuracy is the mean of the three dataset accuracies; parsed is the fraction of all 92 outputs accepted by the symbolic backend. Failed translations remain in the score.

Checkpoint	Macro accuracy	Parsed	Translation time
Qwen3-4B-Instruct-2507	78.61%	84.78%	115.3 s
Qwen3.5-4B	67.50%	53.26%	441.6 s
Qwen3.5-9B	73.33%	70.65%	561.5 s

Qwen/Qwen3-4B-Instruct-2507 was retained. On the same training subset, the rule grammar scored 24/24 and the causal parser 16/20, compared with 21/24 and 13/20 for the selected model alone. The final pipeline therefore uses the grammars where they apply and the model elsewhere.

The complete pipeline scored 78/92 with sampling, 77/92 with greedy decoding, and 77/92 with one compiler-error repair pass. Sampling without repair was retained: temperature 0.7, top-p 0.8, top-k 20, batch size eight and a 2,048-token output limit. Model selection and decoding comparisons used training examples only. Runs used RTX PRO 6000 Blackwell GPUs; the table excludes model loading and downloads. Checkpoint revisions are pinned in the notebook.

Direct logic

The 204 FOLIO validation examples also include reference formulas. Using those formulas directly, the agent scored 199/204 (97.55% binary accuracy, 93.63% three-way), with five malformed inputs and no timeouts. The run took approximately 0.50 s on CPU; no model was loaded. Parse errors count as False in the binary score. The formula and English evaluations use different inputs for the same examples.

Public validation

The English-input evaluation used 128 RuleTaker dev examples (16 per configuration), all 204 FOLIO v0.0 validation examples, and 200 CounterBench examples held out by context group. CounterBench event names are normalized before the SHA-256 group split to prevent renamed contexts crossing splits. FOLIO uses the original public training release and an unfiltered validation mirror. These public splits were also inspected during development. No private test data was available.

Dataset	N	Binary accuracy	Parse errors	Translation	Logic
CounterBench	200	97.50%	0	0.00 s	0.324 s
FOLIO	204	76.47%	38	120.30 s	0.367 s
RuleTaker	128	100.00%	0	0.00 s	0.490 s

Overall binary accuracy was **479/532 (90.04%)**. Only entailed maps to True; refuted, unknown, parsing errors and timeouts map to False. FOLIO three-way accuracy was 50.00%; its always-False binary baseline was 64.71%. All 38 translation failures were on FOLIO. There were no solver timeouts. CounterBench had 20 contradictory-evidence cases, which remain in the score.

The table gives per-dataset timings from an earlier English-input run; controlled parsing in that run took 1.28 s. Translation times are CUDA-synchronized batch measurements divided over their examples; logic time covers formula compilation and solver checks. A later notebook run took 125.20 s (0.235 s per query), including input normalization and model loading, and excluding dataset reading and CSV writing. The RuleTaker result is for the 128-example sample, not the full dev split.

Reasoning traces

Rule example. Young(bob), Kind(bob), Young(x) implies Smart(x), and Smart(x) AND Kind(x) implies Successful(x). The deductions are Smart(bob), then Successful(bob). Z3 finds KB AND NOT Successful(bob) unsatisfiable, giving True. Separate existential witnesses stay separate: exists x P(x) and exists x Q(x) do not entail exists x (P(x) AND Q(x)).

Causal example. B=A, C=NOT B, D=C. With factual A=True, the values are B=True, C=False, D=False. Under do(A=False), the equations give B=False, C=True, D=True. The counterfactual query D is therefore True.

Limitations

Z3 verifies the translated formulas; translation errors can still change the meaning. Quantifier scope, coreference and predicate naming remain failure points. Closed-world negation and default-True causal roots are dataset conventions, so general inputs use classical open-world logic unless another mode is selected. Probabilistic causal entailment checks all positive-probability worlds rather than estimating probabilities. The supplied CSV contains the 532 public validation predictions.

Sources

Qwen3.5-4B; Qwen3.5-9B; Qwen3-4B-Instruct-2507; RuleTaker; CounterBench; FOLIO original release; FOLIO validation mirror; Z3; Lark; LemmInflect; XGrammar